

Vehicle Tracking + Occlusions Research Project

Phase 0 - Pilot Dataset Construction and Event Verification

FROZEN DOCUMENTATION

Prepared for continuation across project chats
28 September 2026

Purpose: preserve the exact Phase 0 decisions, workflow, verified observations, limitations, and frozen outputs before baseline implementation.

1. Research Goal and Role of Phase 0

Frozen research question. Can reliability-aware sequential acquisition of vehicle identity cues improve post-occlusion recovery of a user-selected road motor vehicle among visually similar distractors, while reducing false identity decisions and average inference cost compared with always-on multimodal Re-ID and fixed coarse-to-fine recovery?

Phase 0 objective. Build a small, manually verified pilot set of vehicle occlusion/recovery situations that is sufficient to debug the experimental protocol and later test tracking/recovery baselines. Phase 0 is not the final benchmark and does not require complete coverage of every scenario before implementation begins.

- Select real candidate events using dataset annotations rather than watching entire datasets manually.
- Verify what is actually visible in the original frames, because annotation overlap does not automatically mean true disappearance.
- Record partial occlusion, severe occlusion, reappearance, distractor ambiguity, repeated occlusions, boundary exits, and no-reappearance windows separately.
- Keep uncertain cases provisional instead of turning annotation metadata into unsupported ground truth.

2. Data and Project Organization

Two datasets are available in the project: UA-DETRAC and BDD100K. The manually verified Phase 0 event log currently uses UA-DETRAC. BDD100K is retained for later gap-filling or broader evaluation if needed.

Resource	Phase 0 use
UA-DETRAC images	Primary source for the pilot events reviewed in Phase 0. Each sequence is stored as consecutive image frames.
UA-DETRAC XML annotations	Used to identify target IDs, bounding boxes, annotated occlusion regions, occluder IDs, vehicle type and candidate frame intervals.
BDD100K	Available but not required for the frozen Phase 0 event log; can later fill missing scenarios or provide broader conditions.

Permanent project structure used during Phase 0:

- data/raw/UA-DETRAC/images/
- data/raw/UA-DETRAC/annotations/{train_xml,test_xml,archives}/
- data/raw/BDD100K/
- data/processed/pilot_events/
- src/data_preparation/
- results/visualizations/
- docs/, models/, notebooks/, configs/

The three UA-DETRAC image sequences verified against their XML files were MVI_39311 (1505 frames), MVI_39401 (1385 frames), and MVI_40761 (2030 frames). All three matching XML files were found in the test annotations.

3. Phase 0 Workflow Actually Performed

- 1. Dataset verification:** Confirmed that each selected image folder contained the expected consecutive frames and a matching XML annotation file.
- 2. Automatic occlusion extraction:** Created `find_occlusion_events.py` to parse XML annotations and group annotated occlusion intervals by target.
- 3. First screening:** MVI_39311 initially produced 58 candidate intervals; a simple duration filter yielded 26 manageable candidates. The first three contact sheets showed that annotation duration alone was not a sufficient selection criterion.
- 4. Severity-aware screening:** Updated the extractor to compute target overlap ratio, exact union area of multiple occlusion regions, target size, nearby same-type vehicles, and contextual visibility before/after each interval.
- 5. Three-sequence screening:** The updated script found 71/6 total/shortlisted intervals in MVI_39311, 55/7 in MVI_39401, and 206/78 in MVI_40761 - 332 annotated intervals and 91 initial shortlisted intervals in total.
- 6. Detailed visualization:** Created detailed contact sheets showing full frames and enlarged crops before, during, at peak overlap, and after each candidate.
- 7. Manual verification:** Reviewed original frames for selected targets. Manual observations were treated as the authority for visibility, while XML remained useful for identity and annotation context.
- 8. Gap-directed ranking:** Created `rank_next_candidates.py` to prioritize remaining candidates using target motion, severe occlusion and nearby same-type vehicles, leading to additional review of Targets 39, 35 and 24.
- 9. Freeze:** Consolidated reviewed events into a single `pilot_event_log.csv` and mapped them to `scenario_coverage.csv`. Overconfident labels were corrected before freezing.

4. Phase 0 Scripts and Their Purpose

Script	Purpose
<code>find_occlusion_events.py</code>	Parse UA-DETRAC XML; calculate occlusion severity; group intervals; identify context frames; estimate nearby same-type vehicles; write all-events and shortlist CSVs.
<code>visualize_events.py</code>	Create detailed contact sheets for specific target/interval pairs using full frames, enlarged crops, target boxes, occluders and nearby same-type IDs.
<code>rank_next_candidates.py</code>	Rank remaining candidates by motion, overlap severity and nearby same-type counts, while excluding already reviewed targets.

5. Representative Visual Verification Examples

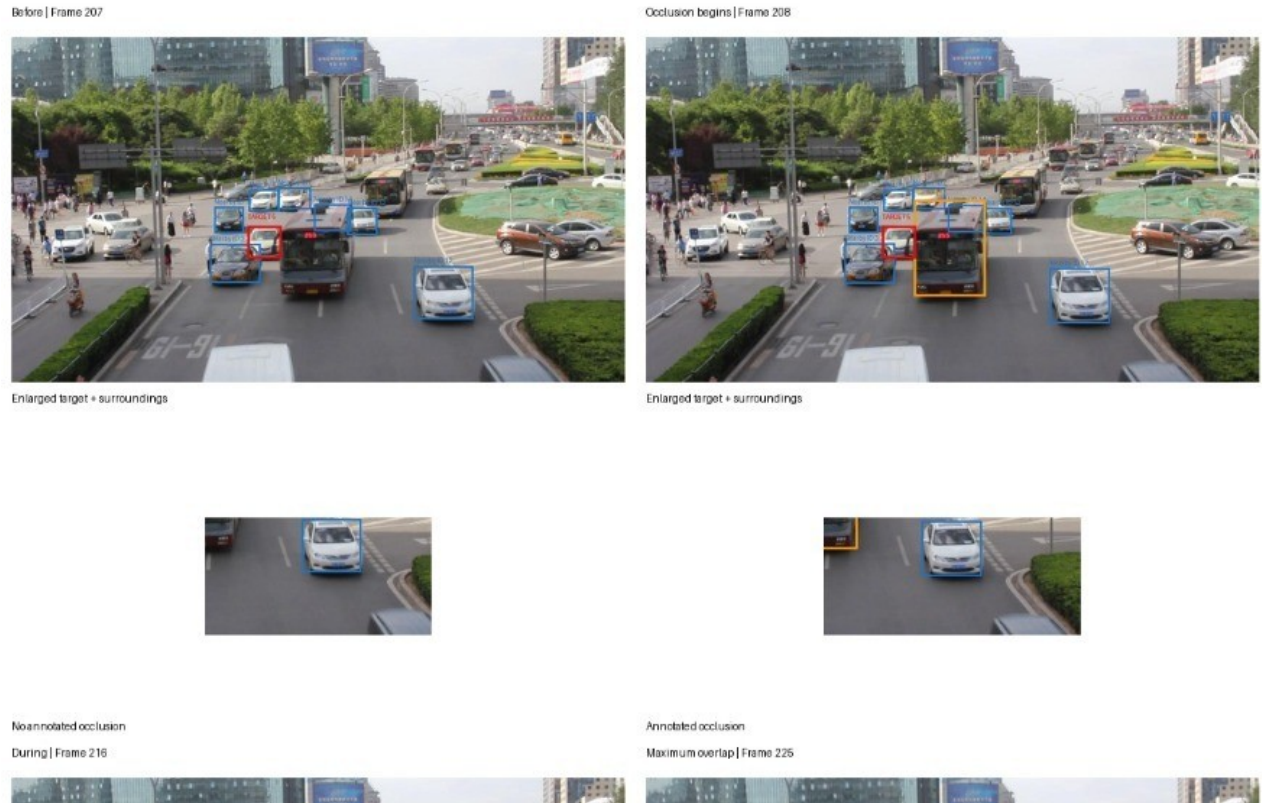


Figure 1. MVI_39311, Target 5: severe partial occlusion by a bus. Manual review showed that a small roof portion remained visible, so this was not treated as a confirmed full disappearance.

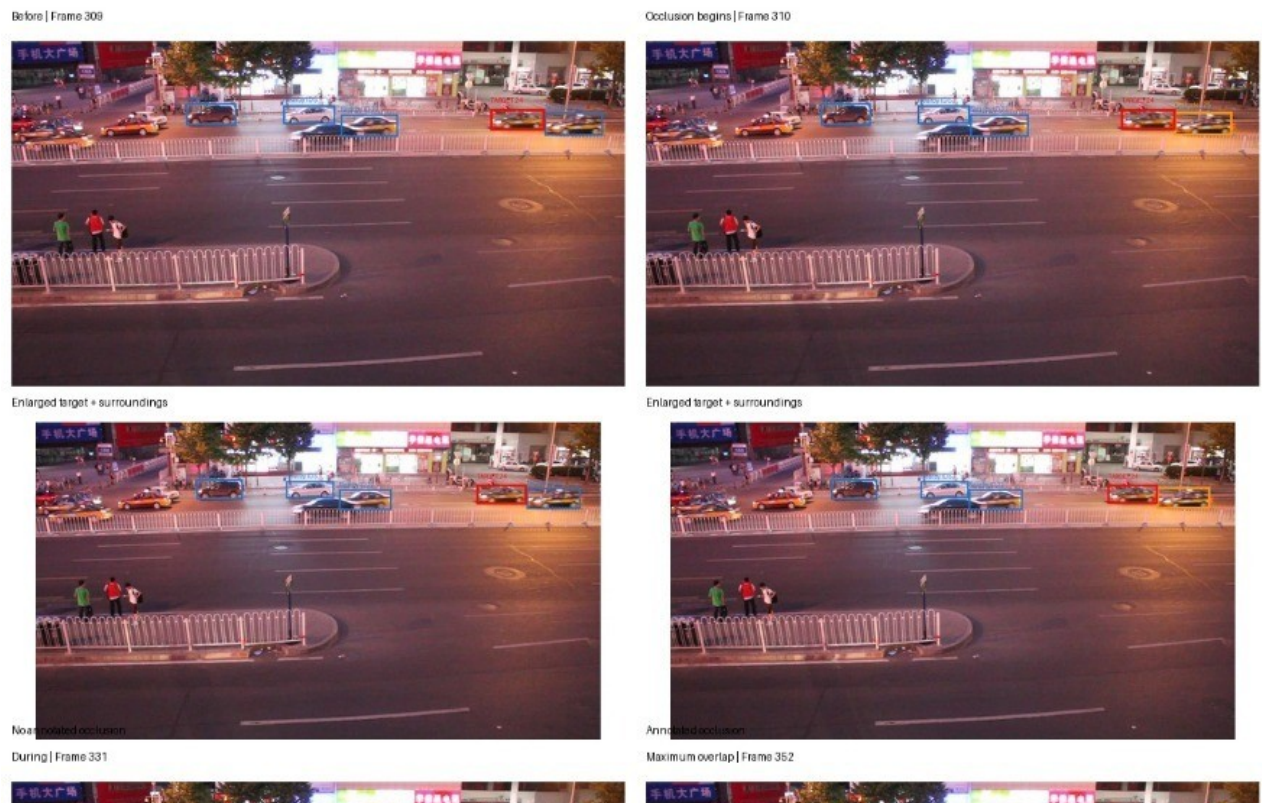


Figure 2. MVI_40761, Target 24: moving taxi interacting with another similar taxi. This became one of the strongest pilot cases for same-type distractor ambiguity.

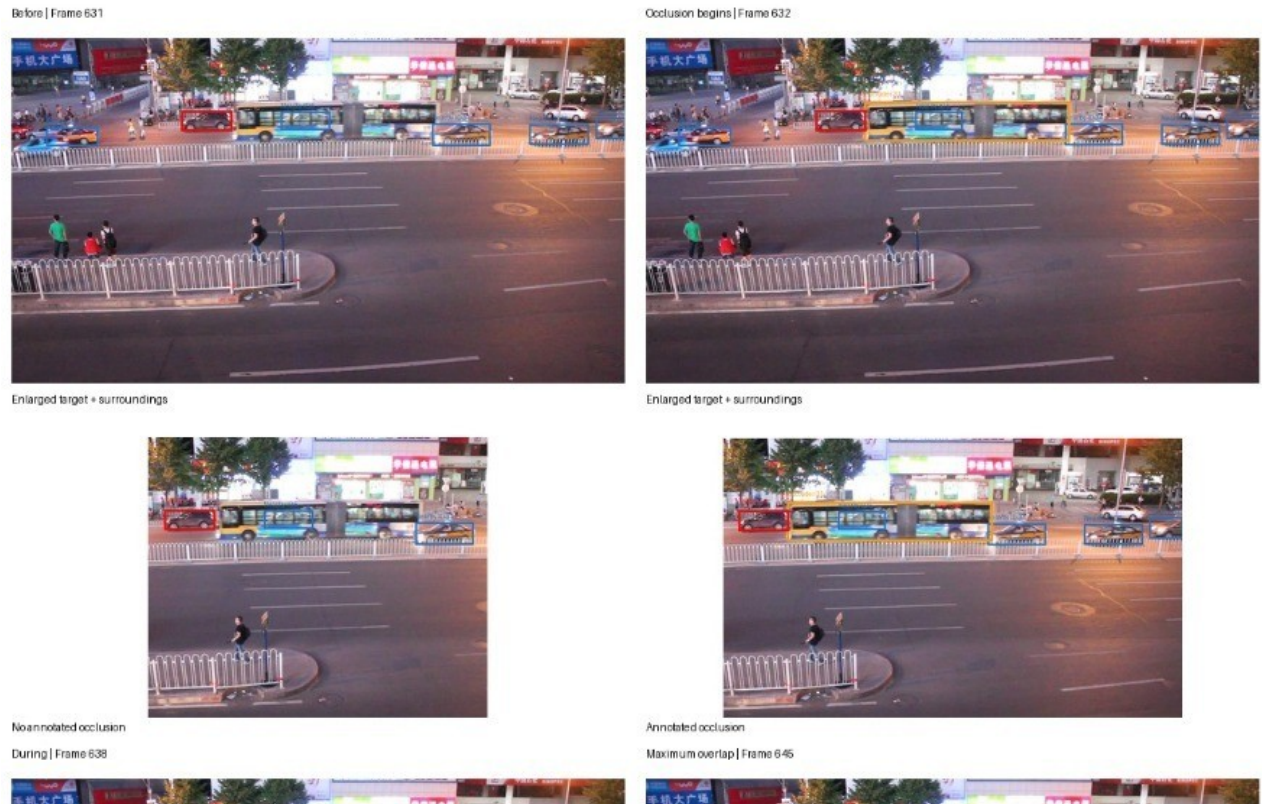


Figure 3. MVI_40761, Target 4: parked black SUV subject to repeated bus occlusions. Useful for repeated/long occlusion, but easier for motion reasoning because the target is stationary.

6. Frozen Pilot Event Log (E001-E012)

The table below is the compact documentation view of the frozen CSV. Frame boundaries marked in the notes are approximate unless explicitly established during manual review.

ID	Sequence / target	Scenario	Key frames	Full loss?	Distractors	Status
E001	MVI_39311 / T5	partial occlusion	start 51; reappear ~172	no	unverified	manually reviewed
E002	MVI_39311 / T5	severe partial occlusion	start 200; max ~225; reappear ~271; clear ~356	no	unverified	manually reviewed
E003	MVI_39401 / T92	full occlusion recovery candidate	start 955; max ~976; reappear ~1016	likely_yes	possible	provisionally selected
E004	MVI_39401 / T75	partial occlusion and boundary exit	start 634; max ~675; reappear ~720	no	possible	manually reviewed
E005	MVI_40761 / T4	repeated full occlusion candidate	start 865; max ~882; reappear ~1063	likely_yes	unverified	manually reviewed
E006	MVI_40761 / T4	repeated full occlusion candidate	start 1084; max ~1098; reappear ~1249	likely_yes	unverified	manually reviewed
E007	MVI_40761 / T4	repeated full occlusion candidate	start 1324; reappear ~1413	likely_yes	unverified	manually reviewed
E008	MVI_39401 / T39	moving target with distractors	start 250; max ~286; reappear ~350; clear ~397	no	yes	manually reviewed
E009	MVI_39311 / T35	moving partial occlusion	start 1043; max ~1075; reappear	no	yes	manually reviewed

			~1113			
E010	MVI_39311 / T35	no reappearance before sequence end	start 1391; max ~1391	unknown	yes	manually reviewed
E011	MVI_40761 / T24	moving similar distractor occlusion	start 315; max ~353; reappear ~378	no	yes	manually reviewed
E012	MVI_40761 / T24	repeated occlusion with similar distractors	start 462; max ~534; clear ~608	no	yes	manually reviewed

7. Event-Level Findings That Matter for Later Experiments

Target 5, MVI_39311 (E001-E002): Two partial/severe partial occlusion episodes. A small part of the vehicle remained visible during the bus occlusion. Useful for testing robustness to partial evidence, not a confirmed full-disappearance case.

Target 92, MVI_39401 (E003): Bus occlusion appears to fully hide the moving target around frame 976; it becomes visible near the image boundary around 1016-1019 and quickly exits. Retained only as a provisional full-occlusion recovery candidate because exact invisible boundaries remain approximate.

Target 75, MVI_39401 (E004): Pole/traffic partial occlusion followed by lane/position change and eventual scene exit. Useful as a partial-occlusion/boundary case, not a main full-recovery example.

Target 4, MVI_40761 (E005-E007): Parked black SUV repeatedly disappears behind buses and reappears. Strong repeated/long occlusion material, but its stationary position makes it easier for motion-based recovery than a moving target.

Target 39, MVI_39401 (E008): Moving black SUV partially blocked by a pole while entering a main road with many dark vehicles. Useful for moving-target distractor ambiguity, but not a full-disappearance case.

Target 35, MVI_39311 (E009-E010): Moving taxi partially occluded by a white car; later, while stationary, a van covers it through the end of the sequence. E010 is a no-observed-reappearance window, not proof that the target is physically absent from the scene.

Target 24, MVI_40761 (E011-E012): Moving taxi interacts with other visually similar taxis. One taxi catches up and obscures it while parallel; later the target turns near a stopped taxi and is heavily covered before exiting the frame. This is the strongest Phase 0 evidence for same-type distractor ambiguity.

8. Frozen Scenario Coverage

ID	Scenario	Status	Events	Phase 0 note
S01	Short occlusion then reappearance	Covered	E001;E002;E011	Multiple partial/severe occlusion cases with reappearance
S02	Same-color/body-type distractors	Covered	E011;E012	Target 24 interacts with visually similar taxis
S03	Nearly identical vehicles together	Provisional	E011;E012	Similar taxis present but near-identical difficulty not formally verified
S04	Unreadable or	Provisional	E003;E011;E012	Targets are too

	unavailable plate			small/distant for reliable plate use; not explicitly measured yet
S05	Changed viewpoint	Provisional	E008;E012	Moving vehicles change position/orientation but no strong viewpoint-specific case yet
S06	Long disappearance	Provisional	E005;E006	E005 and E006 contain long occlusion candidates, but exact invisible-frame boundaries remain approximate and the target is stationary.
S07	Target absent during observation window	Missing	-	E010 has no observed reappearance before sequence end, but the target remains occluded in-scene rather than being verified absent while distractors appear. A true target-absent observation window is still needed for final evaluation.
S08	Distractors appear before target	Missing	-	Need targeted example
S09	Repeated occlusions	Covered	E005;E006;E007	Same target repeatedly hidden and reappearing
S10	Lighting/resolution changes	Provisional	E005;E006;E007;E011;E012	Pilot includes daytime and nighttime sequences, but no verified within-event lighting or resolution transition. Treat as dataset-level condition coverage only.

Coverage interpretation. Phase 0 is intentionally not a complete benchmark. S01, S02 and S09 have direct pilot coverage; several other scenarios are provisional; S07 and S08 remain missing in their strict definitions. This does not block baseline implementation. Missing or weak scenarios should be collected later only when they are needed for final evaluation or exposed by baseline failure analysis.

9. Methodological Corrections Made Before Freeze

- **Annotated overlap is not identical to visual disappearance:** A 100% XML overlap ratio can indicate that the target box is fully covered by annotated overlap, but it does not independently prove that the target is visually unobservable. Original frames remain necessary.
- **Same vehicle type is not the same as visual similarity:** The XML vehicle_type field was used only as a screening proxy. Manual review was required to call vehicles visually similar.
- **Repeated events from one target are not independent samples:** E005-E007 all involve the same parked SUV. They are useful repeated-occlusion episodes but must not be treated as three independent vehicles.

- **Stationary-target recovery is easier for motion reasoning:** Target 4 is useful for occlusion stress testing but should not be used alone as evidence for difficult moving-vehicle identity recovery.
- **No observed reappearance is not the same as target absence:** E010 ends while the target is still occluded. S07 therefore remains missing under the strict definition: target absent from the observation window while distractors appear.
- **Test split reuse must be controlled:** The three repeatedly inspected UA-DETRAC sequences are development/pilot material. Final experiments should reserve untouched sequences so that tuning decisions are not evaluated on the same examples.

10. Frozen Phase 0 Outputs

Artifact	Role
pilot_event_log_phase0_frozen.csv	Authoritative Phase 0 record of 12 manually reviewed/provisional events (E001-E012).
scenario_coverage_phase0_frozen.csv	Scenario-level summary for S01-S10, including covered, provisional and missing cases.
*_all_events.csv	Automatic XML-derived interval inventories for each screened sequence.
*_shortlist.csv	Severity/size/context-screened candidate intervals.
Detailed contact sheets	Visual evidence used for manual target identification and event review.

11. Phase 0 Freeze Decision and Next Stage

PHASE 0 STATUS: FROZEN. Do not continue collecting random occlusion examples. Reopen Phase 0 only if baseline failures or final experimental design require a missing/weak scenario such as strict target absence (S07), distractors-before-target (S08), a stronger near-identical case (S03), or a stronger viewpoint/condition transition.

Immediate next stage: begin the tracking/recovery baseline. Existing tracking work from the literature review should be reused where suitable rather than rebuilding every component from scratch. A tracker such as DAM4SAM can be evaluated as an existing baseline/component if its setup and compute requirements are practical; it does not by itself solve the complete reliability-aware sequential identity-recovery research question.

- Run a user-selected tracking baseline on frozen pilot events, especially E011 (moving target + similar taxi distractor).
- Measure correct identity continuation, target loss, identity switches, recovery, abstention/reject behavior where applicable, latency and inference cost.
- Use tracker failures to decide which missing scenarios need targeted additional data.
- Continue deep reading by module: tracking/occlusion/recovery first, then vehicle Re-ID, uncertainty/reliability, sequential cue acquisition, multimodal fusion and reject option.
- Keep pilot development data separate from untouched final evaluation sequences.

Appendix A. Phase 0 Numbers at Freeze

Measure	Value
UA-DETRAC sequences deeply inspected	3 (MVI_39311, MVI_39401, MVI_40761)
Frames in those sequences	1505 + 1385 + 2030
Annotated intervals after severity-aware parsing	332
Initial shortlisted intervals	91
Manually logged Phase 0 events	12 (E001-E012)
Scenario definitions tracked	10 (S01-S10)
Scenarios marked covered	S01, S02, S09
Scenarios marked missing	S07, S08
Remaining scenarios	Provisional